One day a boy came with a story.

One → assigning a number {Word based tokanization}

with → {word character based tokanization}

Vocabulary → Token, Token Id

they make a huge input file. Hard for the transformer to read.

dataset → Byte Pair Encoding Algorithm (Tokenier) → Token → Token Id

Intermediate {Subword Tokenization}

Vocabulary
- character (every characters)
- words (common, repeated)
- subwords (rarely used)

Our goal is to do 2 thing

Tokenize data

GPT-2 subword tokenizer (BPE)

tokens like
[442, 518, 213, 456]
[{id: [442, 518, 213, 456], 'len': 5}]

Apply on full data

store all the token IDs in .bin file (cause stores directly on disk)

Create a .bin file

Create a memory mapped array

split dataset into batches and add everything to train.bin

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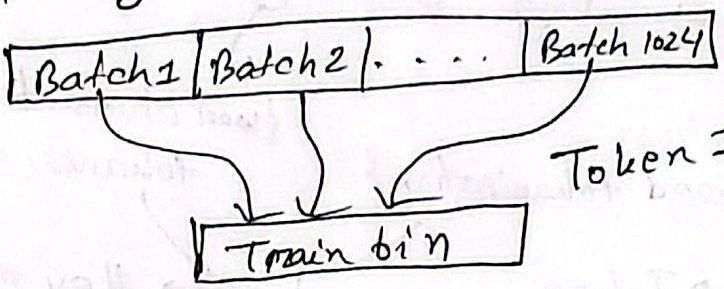
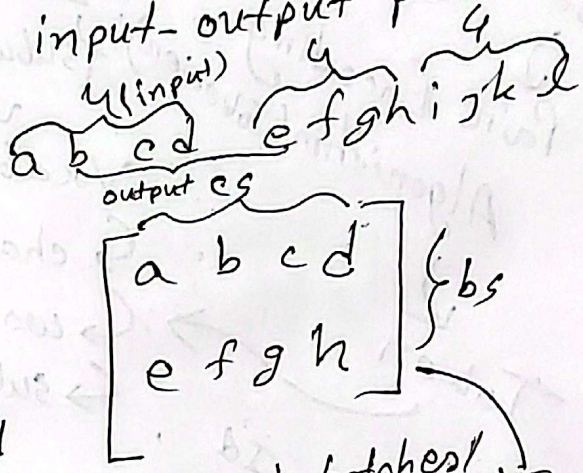
2.12M story → 1024 batch



Token Ids are collected

memory mapped array
or np.memmap()
stores data in disk

Creating input-output pair



context size = 4

batch size = 2

output is input shifted one in right

Input batches/ tensor

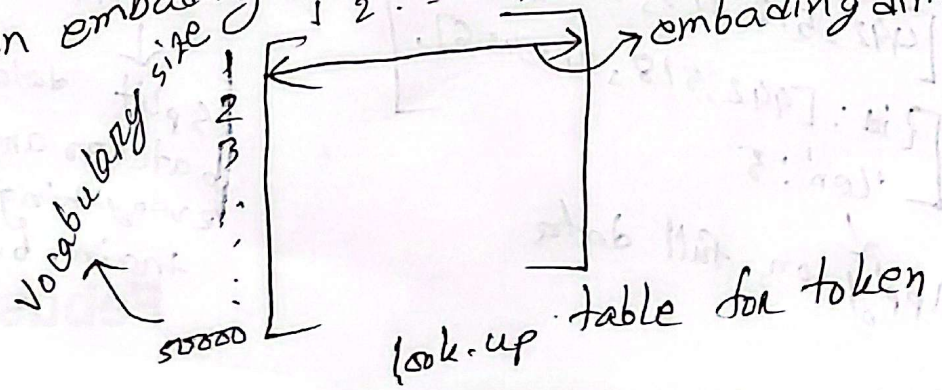
Token = [id: [442, 10, 11, 12], len: 4]

[b c d e
f g h i]

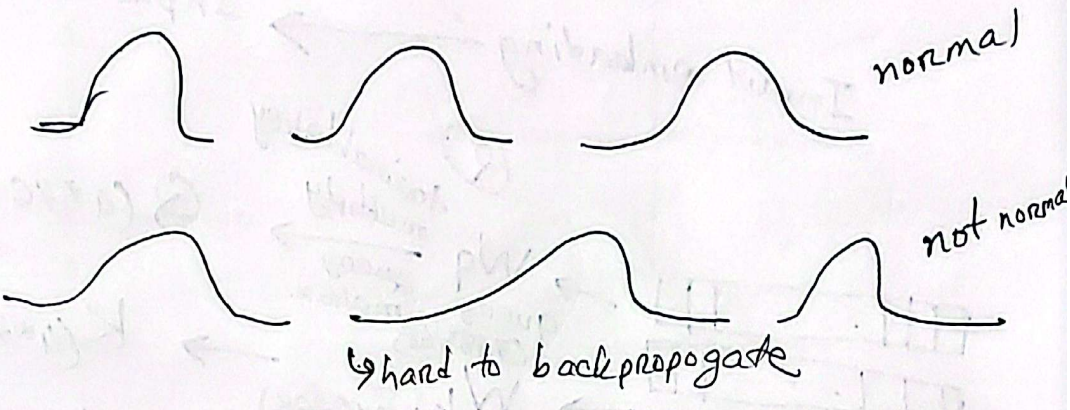
output batches tensor

* converting a token ID into a higher dimension vector is Token Embading → projecting into

Token embading matrix: 268 Embading size



Layer normalization decrease internal covarient shift.

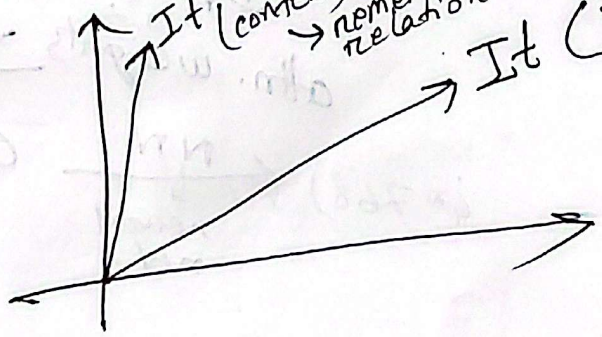


Multi head attention

The dog chased a ball. It could not chase it.

attention augments a value to know about neighbour

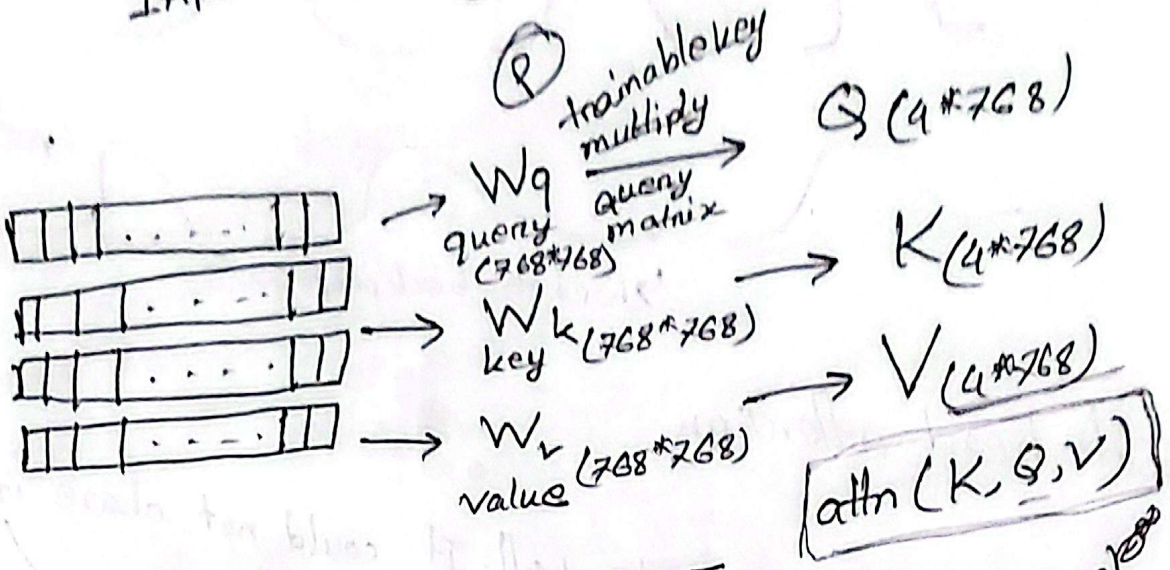
attention ~~or~~ maintain a value as it is. But it creates a context vector of that value. Context correlate with other vector.



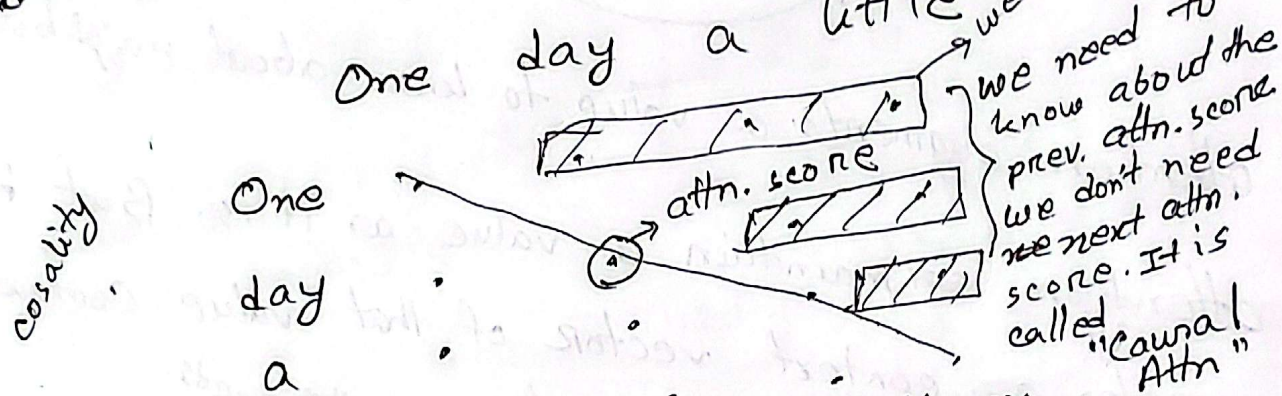
multihead attn. collects the attn. scores with neighbour

After multihead attn.

Input embading → Input context



attn. score matrix : $Q * K^T$



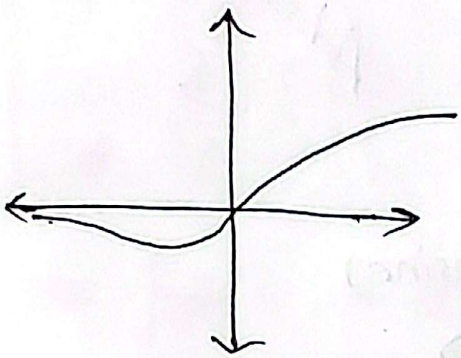
$\frac{1}{\sqrt{dim_k}}$ + softmax

attn. weights → * Values → Context Matrix (4*768)

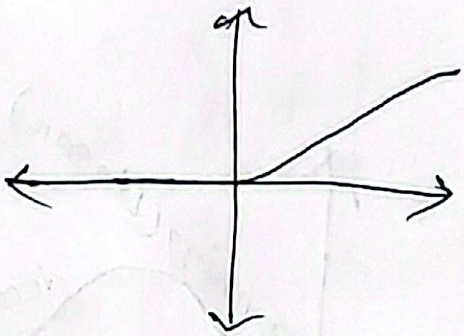
(4*768) ← NN (neural net.)

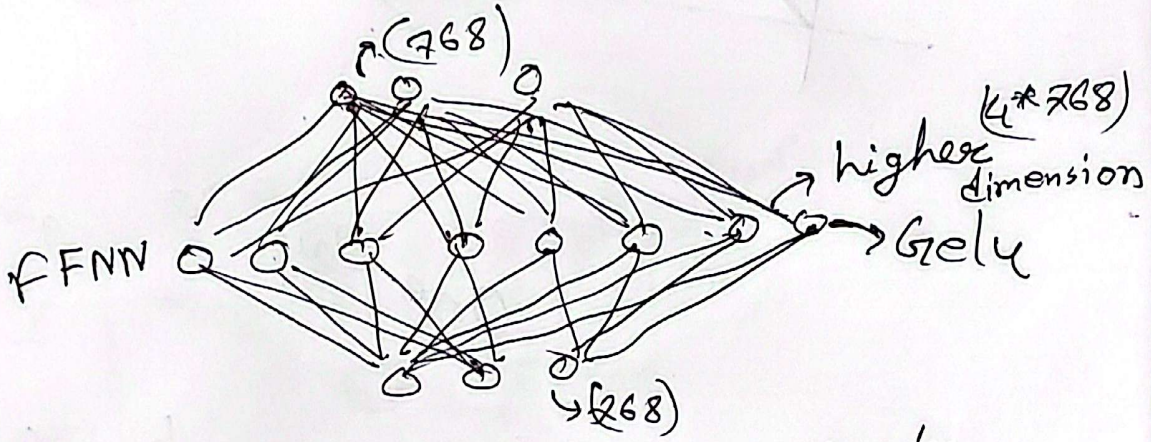
In L2 we keep mean = 0, variance = 1 → feed forward Neural Network

Gelu :

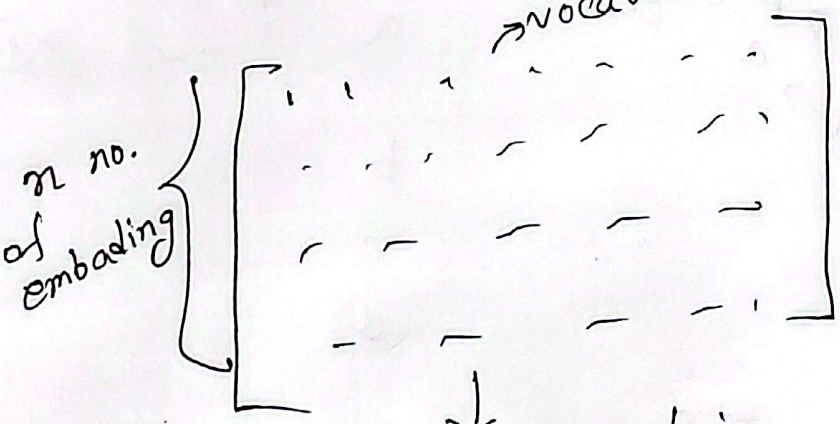


Relu :





Then again dropout randomly.



↓ Logits Matrix

$\begin{bmatrix} p_1 \\ p_2 \\ p_3 \\ p_4 \end{bmatrix}$

↓ Loss

$-\frac{1}{4}[\log p_1 + \log p_2 + \log p_3 + \log p_4]$

↳ negative log likelihood loss

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